Scenario 1: Hospital laboratory test delays

Problem: A hospital's laboratory is taking too long to deliver test results, delaying doctors' treatment decisions.

1. Descriptive — What happened?

The average turnaround time for laboratory results increased from 45 minutes to 90 minutes over two weeks. Nearly 30% of urgent tests missed the hospital's target.

2. Diagnostic — Why did it happen?

Analysis reveals that delays are concentrated between 8 AM and 11 AM. The laboratory receives a surge of samples during this period, while only two technicians are available to process them.

3. Predictive — What might happen next?

If sample volumes continue to increase and staffing remains unchanged, the laboratory is likely to experience longer waiting times during the next morning peak.

4. Prescriptive — What should we do now?

Assign an additional technician during peak hours, prioritize urgent samples, and monitor turnaround time by test type and shift.

Scenario 2: Cloud infrastructure cost spikes

Problem: A software company receives an unexpectedly high cloud computing bill despite having roughly the same number of customers.

1. Descriptive — What happened?

The monthly cloud bill increased by 40%, while the number of active users increased by only 5%.

2. Diagnostic — Why did it happen?

Cost analysis shows that compute instances remained active overnight, development environments were left running, and a background data-processing job repeatedly retried failed tasks.

3. Predictive — What might happen next?

If these resource usage patterns continue, the next month's bill may remain elevated or increase further, particularly as data-processing workloads grow.

4. Prescriptive — What should we do now?

Schedule non-production resources to shut down after working hours, fix the retry loop, right-size oversized instances, and configure budget alerts.

Scenario 3: Food waste in a hotel kitchen

Problem: A hotel prepares food for its buffet every day, but large quantities remain unused and are discarded.

1. Descriptive — What happened?

Kitchen records show that 22% of prepared buffet food was discarded last month. Waste was especially high for breakfast items on weekdays.

2. Diagnostic — Why did it happen?

Comparing food preparation records with guest counts reveals that the kitchen used weekend occupancy estimates to plan weekday quantities. It also prepared large batches before knowing actual guest demand.

3. Predictive — What might happen next?

If the same preparation pattern continues, the hotel is likely to keep wasting food on low-occupancy days, increasing ingredient costs and disposal volumes.

4. Prescriptive — What should we do now?

Forecast meal demand using room occupancy, reservations, and historical consumption. Prepare smaller batches, replenish the buffet as needed, and track waste by dish.

Scenario 4: Railway track maintenance failures

Problem: A railway operator experiences repeated equipment faults and service disruptions on certain track sections.

1. Descriptive — What happened?

Maintenance records show that track-related faults increased over three months, with several sections accounting for most of the reported incidents.

2. Diagnostic — Why did it happen?

Engineers compare fault reports with inspection records, train traffic, weather, and vibration measurements. They discover that the affected sections have elevated vibration levels and recurring drainage problems.

3. Predictive — What might happen next?

A validated predictive-maintenance model may identify track sections with a higher probability of future faults based on vibration trends, previous defects, and inspection data.

4. Prescriptive — What should we do now?

Prioritize engineering inspections of high-risk sections, repair drainage issues, schedule preventive maintenance, and impose appropriate operating restrictions when safety assessments require them.

Scenario 5: Cybersecurity alerts in a company

Problem: A company's security team receives thousands of alerts every day and struggles to identify genuine threats among false alarms.

1. Descriptive — What happened?

The security dashboard shows a sharp increase in login alerts, repeated failed authentication attempts, and alerts involving unusual access locations.

2. Diagnostic — Why did it happen?

Investigation shows that many alerts originate from automated login attempts against employee accounts. Other alerts are caused by legitimate employees using VPNs or travelling, creating false positives.

3. Predictive — What might happen next?

Based on observed attack patterns, the security system may estimate which accounts or systems face an elevated risk of compromise if suspicious activity continues.

4. Prescriptive — What should we do now?

Prioritize alerts using risk scores, enforce multifactor authentication, apply rate limits to repeated login attempts, investigate high-risk accounts, and automate safe containment actions with appropriate safeguards.